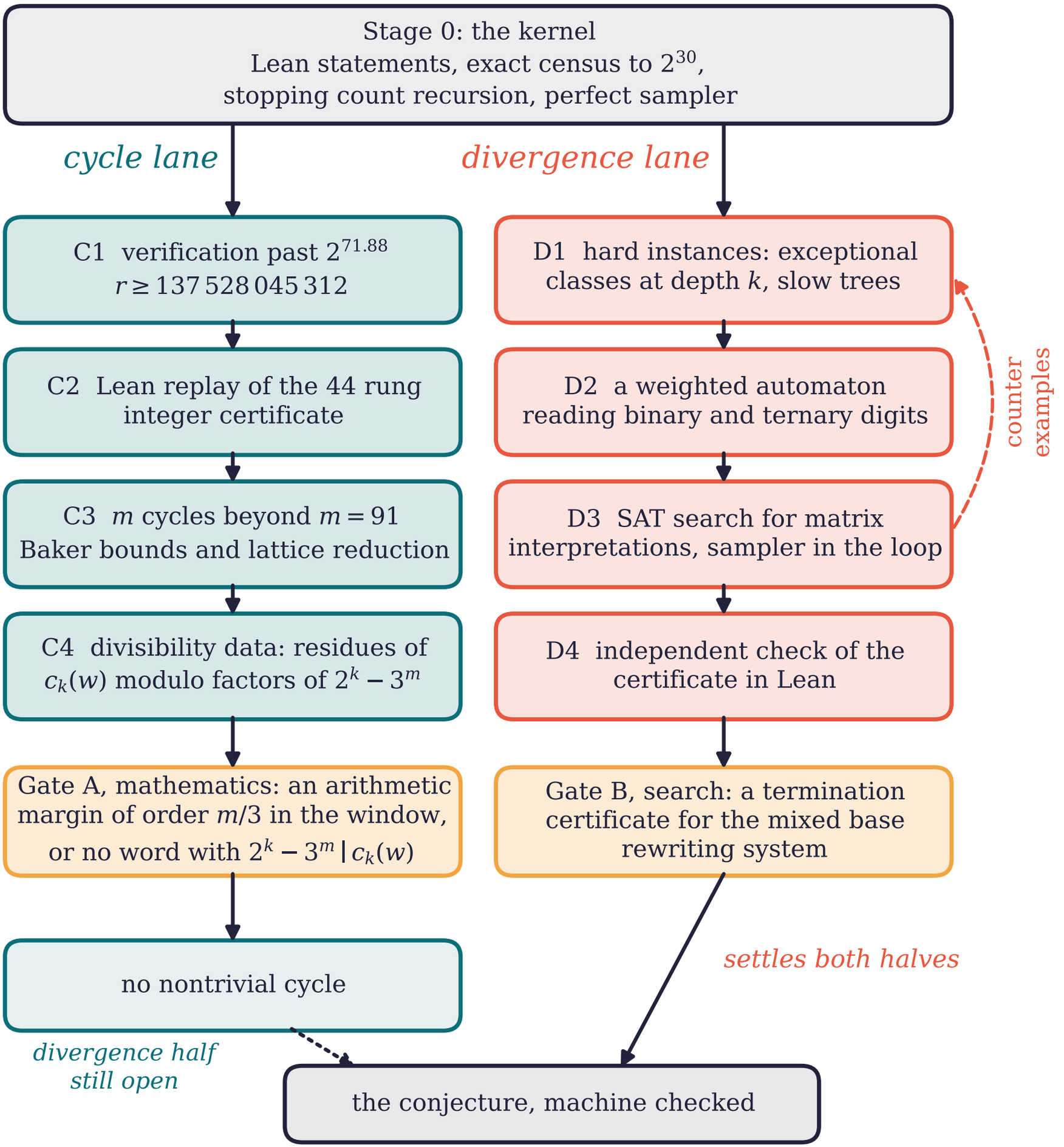
Stage 0: the kernel
Lean statements, exact census to 2^{30},
stopping count recursion, perfect sampler
cycle lane
divergence lane
C1 verification past $2^{71.88}$
$r \geq 137\,528\,045\,312$
C2 Lean replay of the 44 rung
integer certificate
C3 m cycles beyond $m = 91$
Baker bounds and lattice reduction
C4 divisibility data: residues of
$c_k(w)$ modulo factors of $2^k - 3^m$
Gate A, mathematics: an arithmetic
margin of order $m/3$ in the window,
or no word with $2^k - 3^m \mid c_k(w)$
no nontrivial cycle
divergence half
still open
D1 hard instances: exceptional
classes at depth k, slow trees
D2 a weighted automaton
reading binary and ternary digits
D3 SAT search for matrix
interpretations, sampler in the loop
D4 independent check of the
certificate in Lean
Gate B, search: a termination
certificate for the mixed base
rewriting system
counter
examples
settles both halves
the conjecture, machine checked